Digital Image Processing — Team Project Proposal

Restoring Visibility Through Haze, Water, Glass and Plastic: A Comparative Study of Classical Methods, with the Dark Channel Prior (DCP) as the Baseline

Noel Jacob Shaik Areesh Sri Harshit K

CS7.404 Digital Image Processing, Monsoon 2026
October 1, 2026

Abstract

Camera visibility is degraded either by the medium between the scene and the sensor (atmospheric haze, turbid water) or by a protective or intervening element in front of the lens (fogged or dirty glass covers, translucent or aged plastic). This project asks which classical, non-learning image-processing methods restore visibility best in each case, why they succeed or fail, and how far a synthetically generated degradation can stand in for a real capture. The Dark Channel Prior (DCP) is used throughout as the reference method and as the lens for explaining results; no machine learning is used in any restoration or degradation step.

A real, self-captured dataset across the four media is compared against a calibrated synthetic-degradation model (single-layer and stacked multi-layer), with DCP plus a small set of classical and medium-aware alternatives run on both. The deliverable list is kept lean – ten items – by grouping closely related work under a single heading. Three tracks (real data, synthetic data, restoration) run in parallel and converge on a single comparison and analysis spine, mapped against the course's own Report 1 / Report 2 / presentation dates.

Contents

1 Problem Statement and Objectives **1**

2 Data: Real, Synthetic and Existing Image Sets **2**

3 Pipeline: Hand-Implemented vs. Library **3**

4 Deliverables: Order and Dependencies **4**

5 Work Division **5**

6 Timeline Against Course Dates **6**

7 Cut List and Open Items **6**

1 Problem Statement and Objectives

Camera visibility is degraded either by the medium between the scene and the sensor (atmospheric haze, turbid water) or by a protective or intervening element in front of the lens (fogged or dirty

glass covers, translucent or aged plastic). These degradations differ physically, but all reduce contrast and visibility in similar ways. This project asks which classical, non-learning image-processing methods restore visibility best in each case, why they succeed or fail, and how far a synthetically generated degradation can stand in for a real capture. The Dark Channel Prior (DCP) is used throughout as the reference method and as the lens for explaining results. No machine learning is used in any restoration or degradation step.

- Implement the Dark Channel Prior from scratch as the baseline restoration method.
- Build a real, self-captured dataset with clear references across four media (haze, water, glass, plastic) at three severity levels.
- Compare DCP against classical and medium-aware alternatives on every medium.
- Build a synthetic degradation model (single-layer and stacked multi-layer) and validate it against real captures.
- Explain successes and failures using dark-channel statistics, transmission behaviour and colour-cast analysis.
- Recommend a simple, medium-aware restoration pipeline for each medium, backed by the results.

2 Data: Real, Synthetic and Existing Image Sets

Real captures are the primary evidence. Matched synthetic images (degrading our own clear references with the forward model, then comparing against the real degraded capture of the same scene) validate that model. Bulk synthetic data, generated from public depth datasets, drives severity sweeps and stacked-layer experiments that would be impractical to capture by hand.

Medium	Real (own captures)	Synthetic (approximation model)	Existing online image sets
Haze	Outdoor haze/fog shots plus a clear reference on a clear day	$t = \exp(-\beta d)$; A near white	O-HAZE / I-HAZE (real haze, paired GT); RESIDE SOTS (paired); RESIDE ITS / NYU Depth v2 (depth, for synthesis)
Water	Aquarium/tub with a reference shot; milk drops for turbidity levels	Per-channel $t_c = \exp(-\beta_c d)$, $\beta_R > \beta_G > \beta_B$; blue-green A	UIEB for real underwater images (pseudo-reference only; rely on no-reference metrics)
Glass	Fog by steam/spray, dirt by dust/smudges, on a clear pane	Near-uniform low-frequency veil (blurred noise), droplet patterns, alpha-blended smudge masks	None fit – captures and synthesis carry this medium
Plastic	Clear, translucent and aged plastic sheets over a scene	Uniform low t, tint gains, Gaussian blur	None fit – captures and synthesis carry this medium
Stacked	Water/haze scene viewed through a fogged or dirty pane	Two-layer model: $I_2 = Jt_1t_2 + A_1t_2(1 - t_1) + A_2(1 - t_2)$	—

Table 1: Data tiers by medium. Real captures are the primary evidence; matched and bulk synthetic data extend coverage where capture is impractical.

Caveat. Synthetic haze generated with the same model DCP inverts favours DCP by construction, so the synthetic-real gap is reported explicitly rather than hidden. Synthesis parameters are calibrated on a few scenes and evaluated on held-out ones. Depth for our own scenes comes from measured distances, a planar ramp, or 2–3 hand-defined layers – no learned depth model is used.

3 Pipeline: Hand-Implemented vs. Library

The rule: anything under study or central to the project's claims is implemented by hand; generic, uncontested primitives use a library. Library use for baselines should be confirmed with the instructor.

Stage	By hand	Library
Dark channel (patch-min over channels)	Yes – pipeline logic is ours	`scipy.ndimage` or `cv2.erode` (cross-check)
Atmospheric light, raw transmission, recovery (t_0 floor)	Yes	—
Guided filter refinement	Optional manual version	`cv2.ximgproc.guidedFilter` (opencv-contrib)
Soft matting (stretch goal)	—	`pymatting`
Medium-aware variants: UDCP/red channel, per-channel or local A, large-patch/near-global t	Yes – modifications of our DCP	—
Gray-world / simple white balance	Manual gains	`cv2.xphoto` as cross-check
Homomorphic filtering (glass)	Yes – log, FFT, high-pass, exp	NumPy FFT
Wiener deblur (plastic)	PSF estimation manual	`skimage.restoration.wiener`
CLAHE, histogram equalization, gamma	—	OpenCV
Retinex (MSR, stretch)	Yes – Gaussian-blur based	—
Median / bilateral cleanup	—	OpenCV
Forward degradation models (all media, stacked)	Yes	NumPy / SciPy for noise & blur helpers
PSNR, SSIM, CIEDE2000	—	`skimage.metrics`, `skimage.color`
UCIQE, entropy, contrast, dark-channel histograms, model-fit residual, ranking correlation	Yes – short formulas, part of the analysis	`scipy.stats` for Spearman/Kendall
Plots and tables	—	Matplotlib, pandas

Table 2: Implementation ownership by pipeline stage.

Generalized skeleton with per-medium presets: preprocess (none / white balance) → medium-light estimate (global / per-channel / local) → dark-channel variant → transmission (raw, then guided filter) → recover → postprocess (CLAHE or contrast stretch). A config file selects the modules per medium. CAP is excluded because its coefficients were fit by supervised learning.

Medium	Variants tested
Haze	DCP, DCP + guided filter, Retinex, CLAHE
Water	White balance then DCP, UDCP, CLAHE
Glass	Large-patch/near-global t, local A, homomorphic filtering, CLAHE
Plastic	White balance then DCP, Wiener deblur, Retinex, CLAHE

Table 3: Medium-aware variant selection.

4 Deliverables: Order and Dependencies

The deliverable list is kept lean – ten items instead of a longer breakdown – by grouping closely related work (for example, capture protocol with the dataset it produces, and the forward-model spec with the code and calibration that implement it) under a single deliverable. Three tracks run in parallel – real data (D1), synthetic data (D2–D3) and restoration (D4–D6) – and converge at D7. From D7 onward the pipeline is strictly serial through to the final deliverables.

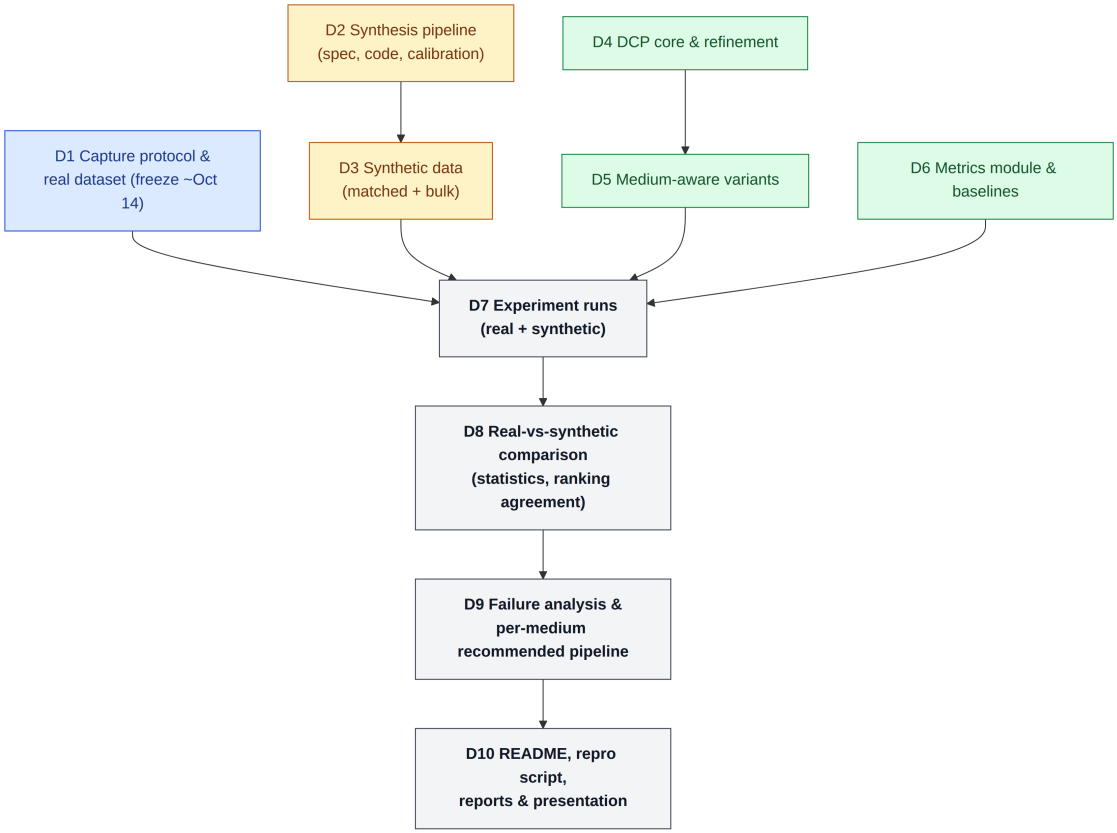


Figure 1: Deliverable dependency graph. Blue = real-data track, amber = synthetic-data track, green = restoration track, grey = the serial spine from experiment runs to final deliverables.

Runs in parallel: D1, D2→D3, and D4→D5 alongside D6, are three independent tracks until they all feed D7. Restoration code (track B) is tested on O-HAZE / RESIDE images while it waits for real and synthetic data to land.

Strictly serial: D1 (freeze) and D3 → D7 → D8 → D9 → D10.

Deliverable	Status	Why
D4 DCP core & refinement, D5 medium-aware variants	**Green**	Well-documented algorithm, small code surface
D6 Metrics & baselines	**Green**	Mostly library calls
D2–D3 Synthesis pipeline & synthetic data	**Amber**	Matching real captures needs tuning
D8 Real-vs-synthetic comparison	**Amber**	Depends on both data tracks landing on time
D9 Failure & stacked-layer analysis	**Amber**	Needs extra captures
Soft matting & Wiener tuning (within D5)	**Amber**	Slow / fiddly – first to cut if behind
D1 Real dataset	**Red**	Capture logistics, lighting and tripod consistency – main schedule risk

Table 4: Risk assessment (RAG status) by deliverable.

5 Work Division

Member	Owns	Deliverables
A – Data & captures	Capture protocol, all real scenes and severity control, depth setups, dataset organisation, downloads (O-HAZE, UIEB, ITS/NYU)	D1, later stacked real shots, demo images
B – Restoration & evaluation	DCP from scratch, refinement, medium-aware variants, baselines, metrics module, experiment runner	D4–D7, D9
C – Synthesis & comparison	Forward models, calibration, matched & bulk synthetic data, real-vs-synthetic statistics and ranking agreement	D2, D3, D8, stacked-layer experiment
All	Understand and be able to explain the DCP core; report sections; rehearsal	D10

Table 5: Work division across the three team members.

A few hours per week per person is realistic for this scope, provided the real dataset is frozen early (around Oct 14) so the busiest coursework weeks are not competing with capture logistics.

6 Timeline Against Course Dates

Dates	Milestone
Sep 28 – 30	Report 1: framing, pipeline diagram, capture protocol, one or two DCP outputs
Oct 1 – 14	Captures (haze & glass, then water & plastic), DCP + refinement, synthesis code. Data frozen around Oct 14
Oct 15 – 26	Variants, baselines, metrics, calibration, first real-vs-synthetic comparison. Report 2 on Oct 26
Oct 27 – Nov 8	Ranking agreement, failure & stacked-layer analysis, severity sweeps, recommended pipelines, README
Nov 9 – 12	Rehearsal. Presentation on Nov 12 (Nov 9 is a holiday – Diwali)

Table 6: Timeline mapped to CS7.404 course milestones.

Assign 3 (due Oct 8), Quiz 2 (Oct 22) and Assign 4 (due Nov 2) overlap this plan, which is why the real dataset freezes before the busiest coursework stretch.

7 Cut List and Open Items

Cut in this order if behind schedule: soft matting → Retinex → plastic stacked-layer cases → bulk severity sweeps.

Must survive: real captures, DCP plus 2–3 classical alternatives, matched synthetic data, and the ranking-agreement comparison.

To confirm with the TA / instructor:

- Whether library calls (guided filter, CLAHE, Wiener, metrics) are acceptable for baselines.
- Whether the reframing from a DCP-robustness study to a multi-method, multi-medium comparison is acceptable, since the submitted proposal was DCP-centric.
- Exact Report 1 requirements.